WORKED SOLUTION

STEP II 2025, Question 5

Pure • Integration by Substitution

STEP II 2025, Question 5

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

Every integral here runs from 0 to ∞, and a substitution of the form $x = u^{-m}$ maps $[0, \infty)$ onto itself, with the ends swapped. The minus sign from $\mathrm{d}x$ undoes the swap of limits, so we land on another integral over $[0, \infty)$. If the new integrand is minus the old one, the integral equals its own negative and so is zero. If it is a multiple of a different integrand of the same shape, we get a relation between two integrals that we never evaluate. In parts (iii) and (iv) we keep the exponent general and choose it so that the powers of u cancel.

Solution

Throughout, if $x = u^{-m}$ with $m > 0$, then $x \to 0$ as $u \to \infty$ and $x \to \infty$ as $u \to 0$, so

$$\int_0^\infty f(x)\,\mathrm{d}x = \int_\infty^0 f(u^{-m})\,(-mu^{-m-1})\,\mathrm{d}u = \int_0^\infty f(u^{-m})\,mu^{-m-1}\,\mathrm{d}u.$$

We use this with $m = 1$ and $m = 2$, and then with general m.

Part (i)

Call the integral I and put $x = u^{-1}$, so $\mathrm{d}x \to u^{-2}\,\mathrm{d}u$ once the limits are put back in order. For $u > 0$,

$$x(x^3+1) = \frac{1}{u}\cdot\frac{1+u^3}{u^3} = \frac{u^3+1}{u^4}, \qquad \text{so} \qquad \sqrt{x(x^3+1)} = \frac{\sqrt{u^3+1}}{u^2},$$

and

$$\sqrt{x} - 1 = \frac{1}{\sqrt{u}} - 1 = \frac{1-\sqrt{u}}{\sqrt{u}}.$$

Hence

$$I = \int_0^\infty \frac{1-\sqrt{u}}{\sqrt{u}} \cdot \frac{u^2}{\sqrt{u^3+1}} \cdot \frac{1}{u^2}\,\mathrm{d}u = \int_0^\infty \frac{1-\sqrt{u}}{\sqrt{u(u^3+1)}}\,\mathrm{d}u$$
$$= -\int_0^\infty \frac{\sqrt{u}-1}{\sqrt{u(u^3+1)}}\,\mathrm{d}u = -I.$$

The last integral is I again, since the name of the variable doesn't matter. So $2I = 0$, giving $I = 0$, as required.

Note that this does not say the integrand is zero. It is negative on $(0,1)$ and positive on $(1,\infty)$, and the substitution matches the two areas exactly.

Part (ii)

Now put $x = u^{-2}$, so $\mathrm{d}x \to 2u^{-3}\,\mathrm{d}u$ after reordering the limits. For $u > 0$,

$$x^3+1 = u^{-6}+1 = \frac{1+u^6}{u^6}, \qquad \sqrt{x^3+1} = \frac{\sqrt{u^6+1}}{u^3}.$$

Therefore

$$\int_0^\infty \frac{\mathrm{d}x}{\sqrt{x^3+1}} = \int_0^\infty \frac{u^3}{\sqrt{u^6+1}} \cdot \frac{2}{u^3}\,\mathrm{d}u = 2\int_0^\infty \frac{\mathrm{d}u}{\sqrt{u^6+1}},$$

which is the required result once we rename u as x.

Part (iii)

Part (i) is the case $s = 1$, $p = 3$, $r = \frac{1}{2}$, so we repeat its method with letters. Call the integral J and put $x = u^{-1}$:

$$x^s(x^p+1) = u^{-s} \cdot \frac{1+u^p}{u^p} = \frac{u^p+1}{u^{s+p}}, \qquad \text{so} \qquad \sqrt{x^s(x^p+1)} = u^{-(s+p)/2}\sqrt{u^p+1},$$

and also $x^r - 1 = (1-u^r)/u^r$. So

$$J = \int_0^\infty \frac{(1-u^r)\,u^{-r}\,u^{(s+p)/2}\,u^{-2}}{\sqrt{u^p+1}}\,\mathrm{d}u = -\int_0^\infty \frac{(u^r-1)\,u^{(s+p)/2-r-2}}{\sqrt{u^p+1}}\,\mathrm{d}u.$$

Compare this with J itself, written as

$$J = \int_0^\infty \frac{(u^r-1)\,u^{-s/2}}{\sqrt{u^p+1}}\,\mathrm{d}u.$$

Ask yourself Why is it enough to match the powers of u, rather than to evaluate anything?

If the two powers of u agree, the substitution turns J into $-J$, and then $J = 0$ exactly as in part (i). So we need

$$\frac{s+p}{2} - r - 2 = -\frac{s}{2} \quad \Longleftrightarrow \quad r = s + \frac{p}{2} - 2.$$

With this value of r we get $J = -J$, so $J = 0$. (The question tells us that p and s are such that the integrals converge, and that is all the argument $J = -J \Rightarrow J = 0$ needs.) As a check, $s = 1$ and $p = 3$ give $r = 1 + \frac{3}{2} - 2 = \frac{1}{2}$, which is part (i).

Part (iv)

Part (ii) is the case $k = 2$, $p = 3$, $q = 6$, which came from $x = u^{-2}$. So we try $x = u^{-m}$ with $m > 0$, so that $\mathrm{d}x \to mu^{-m-1}\,\mathrm{d}u$: since $x^p + 1 = (1 + u^{mp})/u^{mp}$,

$$\int_0^\infty \frac{\mathrm{d}x}{\sqrt{x^p + 1}} = m \int_0^\infty \frac{u^{mp/2}\, u^{-m-1}}{\sqrt{u^{mp} + 1}}\,\mathrm{d}u.$$

The stray power of u disappears when

$$\frac{mp}{2} = m + 1 \quad \Longleftrightarrow \quad p = 2 + \frac{2}{m},$$

and then the right-hand side is $m \displaystyle\int_0^\infty \frac{\mathrm{d}u}{\sqrt{u^q + 1}}$ with $q = mp = 2m + 2$.

Now take $m = k$. For any $k > 0$, the choice

$$p = 2 + \frac{2}{k}, \qquad q = 2k + 2$$

gives

$$\int_0^\infty \frac{\mathrm{d}x}{\sqrt{x^p + 1}} = k \int_0^\infty \frac{\mathrm{d}x}{\sqrt{x^q + 1}}.$$

Both exponents exceed 2, because $k > 0$. That is exactly the condition for these integrals to converge, since the integrand behaves like $x^{-p/2}$ for large x. So the integrals in our relation exist. With $k = 2$ we recover $p = 3$, $q = 6$ from part (ii), and $k = 1$ gives the trivial case $p = q = 4$.

Checks

- Numerically, $\displaystyle\int_0^\infty \frac{\mathrm{d}x}{\sqrt{x^3 + 1}} \approx 2.8044$ and $\displaystyle\int_0^\infty \frac{\mathrm{d}x}{\sqrt{x^6 + 1}} \approx 1.4022$, exactly half, as part (ii) says.
- The integral in part (i) evaluates to zero to machine precision, even though the negative contribution comes mostly from near $x = 0$ and the positive one is spread thinly over $(1, \infty)$.
- For random admissible p and s, a computer check gave zero for the part (iii) integral with $r = s + \frac{p}{2} - 2$. Changing r by 0.3 gave clearly non-zero values.
- For $k = \frac{1}{2}$ (so $p = 6$, $q = 3$) the relation in part (iv) is part (ii) read backwards, as it should be.

Common slips

- Forgetting that the substitution swaps the limits. The minus sign from $\mathrm{d}x$ and the reversed limits cancel, and losing either one gives a wrong sign in part (i).
- Writing $\sqrt{u^{-s}} = u^{-s}/2$ instead of $u^{-s/2}$ when handling the general powers in part (iii).
- In part (iv), finding one pair (p, q) for a particular k rather than giving p and q as functions of every $k > 0$.

Beyond STEP

These integrals can all be evaluated in closed form. Substituting $t = x^p/(1+x^p)$ turns $\int_0^\infty (1+x^p)^{-1/2}\,\mathrm{d}x$ into $\frac{1}{p}\,\mathrm{B}\left(\frac{1}{p}, \frac{1}{2} - \frac{1}{p}\right)$, where B is Euler's Beta function. Combined with the Gamma function, this gives the ratio in part (iv) explicitly. The trick of a symmetry $x \mapsto 1/x$ that sends an integral to minus itself appears again and again in analysis and in contour integration. For a careful treatment of improper integrals, including the convergence questions we were allowed to skip, see Abbott, *Understanding Analysis*. For many more STEP-style substitution tricks, see Siklos, *Advanced Problems in Mathematics*.

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